

MLOps CI/CD Pipeline Report

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Objective

The main objective of this project is to implement a robust Continuous Integration and Continuous Delivery (CI/CD) pipeline for a machine learning model. This ensures automated testing, packaging, and delivery on every codebase change, preventing regression issues in production.

MLOps CI/CD Pipeline Description

Our workflow is built on GitHub Actions and divided into two core phases:

1. **Continuous Integration (CI):** Triggered on every commit/PR to main. It spins up an environment, installs library dependencies from requirements.txt, and executes unit tests using pytest on our scikit-learn Logistic Regression model.
2. **Continuous Delivery (CD):** Runs successfully after tests pass. It structures a deployment-ready package containing model files and uploads it as a securely zipped build artifact.

Experiment Result

The system evaluated model training runs. The test suite successfully verified that the Logistic Regression model consistently achieves an classification accuracy of $\geq 90\%$ (0.90) on the classic Iris dataset, passing our CI threshold step perfectly.

Conclusion

The implementation of the multi-stage CI/CD workflow ensures that the ML software pipeline remains functional, production-ready, and verified automatically without manual human intervention. Future steps include model registry logging and real-time inference monitoring.